

Executive summary

My objective of exploring a honey pot paired up with the IDS suricata system made it so I was able to see how this combined pair was able to be such a good combo when put together my key finding of this task was that I was able to successfully get the IDS to work and enable my attacking VM to be able to be logged when trying to do a scan on my honeypot trap although there was a huge troubleshooting problem this task I was able to fix the issues after hours of configuration online and forum.

HoneyPot lab design

I setup two virtual machine one is called attacker and one is the normal one that I usually used I also changed the network being host only adapter on the virtual machine the tools I used was Dionaea and Suricata.

Testing methodology

I first had to run both of my virtual machines but had to make sure both are on the host only adapter network to make it much easier to do then after that on the defense one I made sure to have the live log of suricata to be open on one terminal and enable Dionaea on another terminal this let me check both without having to get rid of each one every time if it was on one terminal. I then went and look for my ip address on the defense one and gave it to my attacker one where I used a nmap scan which alerted by suricata fast.log it then was able to capture it. I then did a manual interaction with each exposed service. I opened the json logs to take a screen shot of then also took a screenshot of my attackers side of things.

Analysis

Suricata IDS is a really interesting system that lets you know attacks in real time although the installation of the application and Dionaea was a great annoyance since I had to use docker for it and use nano to change some settings everything worked out in the end I did find out that sometimes the attacks if was performed on a local host will not register it as a attack causing me at first to spend hours trouble shooting but after I was able to clone the virtual machine and change the ip address for it I was able to attack the defending VM correctly.

Conclusion & recommendations

I recommend for most people to use two separate vm instead of trying to do a field lab experiment on the same virtual machine as that will not apply the same result or might have a bunch of issue as I have even if it is to test.

Appendices and evidences

-I deployed my Dionaea on docker below

```
(tonytrankiller@ Tony)-[~]
$ sudo docker run -d --name dionaea \
  -p 21:21 -p 23:23 -p 80:80 -p 443:443 \
  dionetools/dionaea
3b7c8244a8bc65a79accafa21d828d38df8a7f3c4b64db68fe21e54ae018aa

(tonytrankiller@ Tony)-[~]
$ sudo docker top dionaea
CONTAINER ID   NAME      CPU %     MEM %     PID        PPID       C          STIME      TTY        P1          P2          TIME       CMD
tonytra-      20157     20133     0           16:09      ?          00:00:00   /opt/dionaea/bin/dionaea -u dionaea -g dionaea -c /opt/dionaea/etc/dionaea/dionaea.cfg
root          20308     20157     0           16:09      ?          00:00:00   /opt/dionaea/bin/dionaea -u dionaea -g dionaea -c /opt/dionaea/etc/dionaea/dionaea.cfg

(tonytrankiller@ Tony)-[~]
$ sudo ss -tnpt $(sudo docker inspect -f '{{.State.Pid}}' dionaea)
ss: cannot connect to host: No such file or directory

(tonytrankiller@ Tony)-[~]
$ sudo netstat -tlnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:21              0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:80              0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:443             0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:1443            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:1433            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:1335            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:142             0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:153             0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:121             0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:123             0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:180             0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:1900            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:1803            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:1723            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:1433            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:1445            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:1443            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:2135            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:2180            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:2153            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:2142            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:2123            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:2121            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:21900           0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:1803            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:1723            0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:127017          0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:213306          0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:112121          0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:2127017         0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:113306          0.0.0.0:*               LISTEN      20157/dionaea
tcp        0      0 0.0.0.0:211211          0.0.0.0:*               LISTEN      20157/dionaea
tcp6       0      0 :::21                  :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::23                  :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::80                  :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::443                 :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::1443                :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::1433                :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::121                 :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::123                 :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::180                 :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::1900                :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::1803                :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::1723                :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::1433                :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::1445                :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::1443                :::*                   LISTEN      20157/dionaea
tcp6       0      0 :::1900                :::*                   LISTEN      20157/dionaea
```

-I tested each of the exposed service on local host at first before I had to knowledge to do it on a attacking VM.

```
(tonytrankiller@ Tony)-[~]
$ curl -I localhost:80
HTTP/1.1 200 OK
Server: nginx
Content-Type: text/html; charset=ascii
Content-Length: 204
Connection: close

(tonytrankiller@ Tony)-[~]
$ ftp localhost 21
Trying [::1]:21 ...
Connected to localhost.
220 DiskStation FTP server ready.
Name (localhost:tonytrankiller): tony
331 Password required for tony.
Password:
230 User logged in, proceed
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
ftp> exit
221 Goodbye.

(tonytrankiller@ Tony)-[~]
$ telnet localhost 23
Trying ::1...
Connected to localhost.
Escape character is '^J'.

Connection closed by foreign host.

(tonytrankiller@ Tony)-[~]
$ smbclient -L //localhost -N

Sharename      Type            Comment
-----
ADMIN$         Disk            Remote Admin
C$             Disk            Default Share
IPC$           IPC             Remote IPC
Printer        Printer         Microsoft XPS Document Writer

Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to localhost failed (Error NT_STATUS_CONNECTION_REFUSED)
Unable to connect with SMB1 -- no workgroup available
```

-The two images below back to back shows me getting the Dionaea logs

```
—(tonytrankiller@Tony)-[~]
$ sudo docker logs -f dionaea
template/etc' → './etc'
template/etc/dionaea' → './etc/dionaea'
template/etc/dionaea/dionaea.cfg' → './etc/dionaea/dionaea.cfg'
template/etc/dionaea/ihandlers-available' → './etc/dionaea/ihandlers-available'
template/etc/dionaea/ihandlers-available/cmdshell.yaml' → './etc/dionaea/ihandlers-available/cmdshell.yaml'
template/etc/dionaea/ihandlers-available/emu_scripts.yaml' → './etc/dionaea/ihandlers-available/emu_scripts.yaml'
template/etc/dionaea/ihandlers-available/emuprofile.yaml' → './etc/dionaea/ihandlers-available/emuprofile.yaml'
template/etc/dionaea/ihandlers-available/fail2ban.yaml' → './etc/dionaea/ihandlers-available/fail2ban.yaml'
template/etc/dionaea/ihandlers-available/ftp.yaml' → './etc/dionaea/ihandlers-available/ftp.yaml'
template/etc/dionaea/ihandlers-available/hpfeeds.yaml' → './etc/dionaea/ihandlers-available/hpfeeds.yaml'
template/etc/dionaea/ihandlers-available/log_db_sql.yaml' → './etc/dionaea/ihandlers-available/log_db_sql.yaml'
template/etc/dionaea/ihandlers-available/log_incident.yaml' → './etc/dionaea/ihandlers-available/log_incident.yaml'
template/etc/dionaea/ihandlers-available/log_json.yaml' → './etc/dionaea/ihandlers-available/log_json.yaml'
template/etc/dionaea/ihandlers-available/log_sqlite.yaml' → './etc/dionaea/ihandlers-available/log_sqlite.yaml'
template/etc/dionaea/ihandlers-available/nfq.yaml' → './etc/dionaea/ihandlers-available/nfq.yaml'
template/etc/dionaea/ihandlers-available/p0f.yaml' → './etc/dionaea/ihandlers-available/p0f.yaml'
template/etc/dionaea/ihandlers-available/s3.yaml' → './etc/dionaea/ihandlers-available/s3.yaml'
template/etc/dionaea/ihandlers-available/store.yaml' → './etc/dionaea/ihandlers-available/store.yaml'
template/etc/dionaea/ihandlers-available/submit_http.yaml' → './etc/dionaea/ihandlers-available/submit_http.yaml'
template/etc/dionaea/ihandlers-available/submit_http_post.yaml' → './etc/dionaea/ihandlers-available/submit_http_post.yaml'
template/etc/dionaea/ihandlers-available/tftp_download.yaml' → './etc/dionaea/ihandlers-available/tftp_download.yaml'
template/etc/dionaea/ihandlers-available/virustotal.yaml' → './etc/dionaea/ihandlers-available/virustotal.yaml'
template/etc/dionaea/ihandlers-enabled' → './etc/dionaea/ihandlers-enabled'
template/etc/dionaea/ihandlers-enabled/cmdshell.yaml' → './etc/dionaea/ihandlers-enabled/cmdshell.yaml'
template/etc/dionaea/ihandlers-enabled/emuprofile.yaml' → './etc/dionaea/ihandlers-enabled/emuprofile.yaml'
template/etc/dionaea/ihandlers-enabled/ftp.yaml' → './etc/dionaea/ihandlers-enabled/ftp.yaml'
template/etc/dionaea/ihandlers-enabled/log_sqlite.yaml' → './etc/dionaea/ihandlers-enabled/log_sqlite.yaml'
template/etc/dionaea/ihandlers-enabled/store.yaml' → './etc/dionaea/ihandlers-enabled/store.yaml'
template/etc/dionaea/ihandlers-enabled/tftp_download.yaml' → './etc/dionaea/ihandlers-enabled/tftp_download.yaml'
template/etc/dionaea/services-available' → './etc/dionaea/services-available'
template/etc/dionaea/services-available/blackhole.yaml' → './etc/dionaea/services-available/blackhole.yaml'
template/etc/dionaea/services-available/epmap.yaml' → './etc/dionaea/services-available/epmap.yaml'
template/etc/dionaea/services-available/ftp.yaml' → './etc/dionaea/services-available/ftp.yaml'
template/etc/dionaea/services-available/http.yaml' → './etc/dionaea/services-available/http.yaml'
template/etc/dionaea/services-available/memcache.yaml' → './etc/dionaea/services-available/memcache.yaml'
template/etc/dionaea/services-available/mirror.yaml' → './etc/dionaea/services-available/mirror.yaml'
template/etc/dionaea/services-available/mongo.yaml' → './etc/dionaea/services-available/mongo.yaml'
template/etc/dionaea/services-available/mqtt.yaml' → './etc/dionaea/services-available/mqtt.yaml'
template/etc/dionaea/services-available/mssql.yaml' → './etc/dionaea/services-available/mssql.yaml'
template/etc/dionaea/services-available/mysql.yaml' → './etc/dionaea/services-available/mysql.yaml'
template/etc/dionaea/services-available/pptp.yaml' → './etc/dionaea/services-available/pptp.yaml'
template/etc/dionaea/services-available/printer.yaml' → './etc/dionaea/services-available/printer.yaml'
template/etc/dionaea/services-available/sip.yaml' → './etc/dionaea/services-available/sip.yaml'
template/etc/dionaea/services-available/smb.yaml' → './etc/dionaea/services-available/smb.yaml'
template/etc/dionaea/services-available/tftp.yaml' → './etc/dionaea/services-available/tftp.yaml'
template/etc/dionaea/services-available/upnp.yaml' → './etc/dionaea/services-available/upnp.yaml'
template/etc/dionaea/services-enabled' → './etc/dionaea/services-enabled'
template/etc/dionaea/services-enabled/blackhole.yaml' → './etc/dionaea/services-enabled/blackhole.yaml'
template/etc/dionaea/services-enabled/epmap.yaml' → './etc/dionaea/services-enabled/epmap.yaml'
template/etc/dionaea/services-enabled/ftp.yaml' → './etc/dionaea/services-enabled/ftp.yaml'
template/etc/dionaea/services-enabled/http.yaml' → './etc/dionaea/services-enabled/http.yaml'
template/etc/dionaea/services-enabled/memcache.yaml' → './etc/dionaea/services-enabled/memcache.yaml'
template/etc/dionaea/services-enabled/mirror.yaml' → './etc/dionaea/services-enabled/mirror.yaml'
template/etc/dionaea/services-enabled/mongo.yaml' → './etc/dionaea/services-enabled/mongo.yaml'
template/etc/dionaea/services-enabled/mqtt.yaml' → './etc/dionaea/services-enabled/mqtt.yaml'
template/etc/dionaea/services-enabled/mssql.yaml' → './etc/dionaea/services-enabled/mssql.yaml'
template/etc/dionaea/services-enabled/mysql.yaml' → './etc/dionaea/services-enabled/mysql.yaml'
template/etc/dionaea/services-enabled/pptp.yaml' → './etc/dionaea/services-enabled/pptp'
```

Session Actions Edit View Help

```
'template/etc/dionaea/services-available/blackhole.yaml' → './etc/dionaea/services-available/blackhole.yaml'
'template/etc/dionaea/services-available/epmap.yaml' → './etc/dionaea/services-available/epmap.yaml'
'template/etc/dionaea/services-available/ftp.yaml' → './etc/dionaea/services-available/ftp.yaml'
'template/etc/dionaea/services-available/http.yaml' → './etc/dionaea/services-available/http.yaml'
'template/etc/dionaea/services-available/memcache.yaml' → './etc/dionaea/services-available/memcache.yaml'
'template/etc/dionaea/services-available/mirror.yaml' → './etc/dionaea/services-available/mirror.yaml'
'template/etc/dionaea/services-available/mongo.yaml' → './etc/dionaea/services-available/mongo.yaml'
'template/etc/dionaea/services-available/mqtt.yaml' → './etc/dionaea/services-available/mqtt.yaml'
'template/etc/dionaea/services-available/mssql.yaml' → './etc/dionaea/services-available/mssql.yaml'
'template/etc/dionaea/services-available/mysql.yaml' → './etc/dionaea/services-available/mysql.yaml'
'template/etc/dionaea/services-available/pptp.yaml' → './etc/dionaea/services-available/pptp.yaml'
'template/etc/dionaea/services-available/printer.yaml' → './etc/dionaea/services-available/printer.yaml'
'template/etc/dionaea/services-available/sip.yaml' → './etc/dionaea/services-available/sip.yaml'
'template/etc/dionaea/services-available/smb.yaml' → './etc/dionaea/services-available/smb.yaml'
'template/etc/dionaea/services-available/tftp.yaml' → './etc/dionaea/services-available/tftp.yaml'
'template/etc/dionaea/services-available/upnp.yaml' → './etc/dionaea/services-available/upnp.yaml'
'template/etc/dionaea/services-enabled' → './etc/dionaea/services-enabled'
'template/etc/dionaea/services-enabled/blackhole.yaml' → './etc/dionaea/services-enabled/blackhole.yaml'
'template/etc/dionaea/services-enabled/epmap.yaml' → './etc/dionaea/services-enabled/epmap.yaml'
'template/etc/dionaea/services-enabled/ftp.yaml' → './etc/dionaea/services-enabled/ftp.yaml'
'template/etc/dionaea/services-enabled/http.yaml' → './etc/dionaea/services-enabled/http.yaml'
'template/etc/dionaea/services-enabled/memcache.yaml' → './etc/dionaea/services-enabled/memcache.yaml'
'template/etc/dionaea/services-enabled/mirror.yaml' → './etc/dionaea/services-enabled/mirror.yaml'
'template/etc/dionaea/services-enabled/mongo.yaml' → './etc/dionaea/services-enabled/mongo.yaml'
'template/etc/dionaea/services-enabled/mqtt.yaml' → './etc/dionaea/services-enabled/mqtt.yaml'
'template/etc/dionaea/services-enabled/mssql.yaml' → './etc/dionaea/services-enabled/mssql.yaml'
'template/etc/dionaea/services-enabled/mysql.yaml' → './etc/dionaea/services-enabled/mysql.yaml'
'template/etc/dionaea/services-enabled/pptp.yaml' → './etc/dionaea/services-enabled/pptp.yaml'
'template/etc/dionaea/services-enabled/printer.yaml' → './etc/dionaea/services-enabled/printer.yaml'
'template/etc/dionaea/services-enabled/sip.yaml' → './etc/dionaea/services-enabled/sip.yaml'
'template/etc/dionaea/services-enabled/smb.yaml' → './etc/dionaea/services-enabled/smb.yaml'
'template/etc/dionaea/services-enabled/tftp.yaml' → './etc/dionaea/services-enabled/tftp.yaml'
'template/etc/dionaea/services-enabled/upnp.yaml' → './etc/dionaea/services-enabled/upnp.yaml'
'template/lib' → 'var/lib'
'template/lib/dionaea' → 'var/lib/dionaea'
'template/lib/dionaea/binaries' → 'var/lib/dionaea/binaries'
'template/lib/dionaea/bistreams' → 'var/lib/dionaea/bistreams'
'template/lib/dionaea/fail2ban' → 'var/lib/dionaea/fail2ban'
'template/lib/dionaea/ftp' → 'var/lib/dionaea/ftp'
'template/lib/dionaea/ftp/root' → 'var/lib/dionaea/ftp/root'
'template/lib/dionaea/http' → 'var/lib/dionaea/http'
'template/lib/dionaea/http/root' → 'var/lib/dionaea/http/root'
'template/lib/dionaea/http/template' → 'var/lib/dionaea/http/template'
'template/lib/dionaea/http/template/example' → 'var/lib/dionaea/http/template/example'
'template/lib/dionaea/http/template/example/form.html.j2' → 'var/lib/dionaea/http/template/example/form.html.j2'
'template/lib/dionaea/http/template/nginx' → 'var/lib/dionaea/http/template/nginx'
'template/lib/dionaea/http/template/nginx/autoindex.html.j2' → 'var/lib/dionaea/http/template/nginx/autoindex.html.j2'
'template/lib/dionaea/http/template/nginx/error.html.j2' → 'var/lib/dionaea/http/template/nginx/error.html.j2'
'template/lib/dionaea/printer' → 'var/lib/dionaea/printer'
'template/lib/dionaea/printer/root' → 'var/lib/dionaea/printer/root'
'template/lib/dionaea/printer/root/0' → 'var/lib/dionaea/printer/root/0'
'template/lib/dionaea/sip' → 'var/lib/dionaea/sip'
'template/lib/dionaea/sip/rtp' → 'var/lib/dionaea/sip/rtp'
'template/lib/dionaea/tftp' → 'var/lib/dionaea/tftp'
'template/lib/dionaea/tftp/root' → 'var/lib/dionaea/tftp/root'
'template/lib/dionaea/upnp' → 'var/lib/dionaea/upnp'
'template/lib/dionaea/upnp/root' → 'var/lib/dionaea/upnp/root'
'template/log' → 'var/log'
'template/log/dionaea' → 'var/log/dionaea'
Starting dionaea ...
```

-Logs of Jsn is in the image below


```
(tonytrankiller@Tony)-[~]
$ ftp 192.168.56.102
Connected to 192.168.56.102.
220 DiskStation FTP server ready.
Name (192.168.56.102:tonytrankiller): ^C
```

File System

```
(tonytrankiller@Tony)-[~]
$ curl http://192.168.56.102
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 3.2 Final//EN"><html>
<title>Directory listing for /</title>
<body>
<h2>Directory listing for /</h2>
<hr>
<ul>
<li><a href="..">../</a>
</ul>
<hr>
</body>
</html>
```

```
(tonytrankiller@Tony)-[~]
$ sudo nmap -sV 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-15 17:56 EST
Stats: 0:00:24 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 60.00% done; ETC: 17:56 (0:00:07 remaining)
Stats: 0:00:25 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 60.00% done; ETC: 17:56 (0:00:08 remaining)
```

```
(tonytrankiller@Tony)-[~]
$ smbclient -L //192.168.56.102
Password for [WORKGROUP\tonytrankiller]:
Server did not provide 'target information', required for NTLMv2
gensec_spnego_client_negTokenTarg_step: SPNEGO(ntlmssp) login failed: NT_STATUS_INVALID_PARAMETER
Anonymous login successful
```

Sharename	Type	Comment
ADMIN\$	Disk	Remote Admin
C\$	Disk	Default Share
IPC\$	IPC	Remote IPC
Printer	Printer	Microsoft XPS Document Writer

```
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 192.168.56.102 failed (Error NT_STATUS_CONNECTION_REFUSED)
Unable to connect with SMB1 -- no workgroup available
```

```
(tonytrankiller@Tony)-[~]
$ curl -I http://192.168.56.102
HTTP/1.1 200 OK
Server: nginx
Content-Type: text/html; charset=ascii
Content-Length: 204
Connection: close
```